ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Inequalities

Solve the boundary first, then decide which side survives.

Inequalities · P1 · 04

eliteigcse.com · WhatsApp +20 112 000 9622

Dr Eslam Ahmed · Cairo University Faculty of Engineering

The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 FLIP: Linear Inequality
TRIGGER *solve an inequality with brackets or fractions*

02 SIGN: Quadratic Sign Chart
TRIGGER *quadratic inequality*

03 SHADE: Graph Region
TRIGGER *region satisfies inequalities*

04 RANGE: Read The Range
TRIGGER *find possible values*

BANK EVIDENCE Local pages: 34 problem, 59 answer, 12 notes. Website primary entries: 5.

01 FLIP: Linear Inequality

TRIGGER *solve an inequality with brackets or fractions*

BECOMES A normal equation with one sign rule.

FIRST LINE TO WRITE

$$ax + b < c$$

SIMPLEST STRATEGY

1. **F**ree brackets and fractions.
2. **L**ine up x-terms on one side.
3. **I**nvert the sign only when multiplying or dividing by a negative.
4. **P**resent the final interval.

WORKED MODEL

$$-2x + 5 < 11 \Rightarrow -2x < 6 \Rightarrow x > -3.$$

HARD VARIANTS

1. Clear fractions carefully; multiply by a negative only when its sign is known.
2. If the multiplier contains x, do not multiply across until you know its sign.
3. Write the final answer as an interval and check one test value.

BOTTOM LINE

Only a negative multiply/divide flips the inequality sign.

Practice 01

SAME IDEA Use FLIP: Linear Inequality.

QUESTION

Solve $7 - 3x \leq 16$.

YOUR SOLUTION

02 SIGN: Quadratic Sign Chart

TRIGGER *quadratic inequality*

BECOMES A number line split by the roots.

FIRST LINE TO WRITE

$$ax^2 + bx + c = 0$$

SIMPLEST STRATEGY

1. **S**olve the boundary equation.
2. **I**nsert roots on a number line.
3. **G**et the sign of each interval.
4. **N**ame the interval that matches the inequality.

WORKED MODEL

$$(x-1)(x-4) < 0 \Rightarrow 1 < x < 4.$$

HARD VARIANTS

1. Repeated roots only touch the axis; the sign may not change there.
2. If the leading coefficient is negative, the outside/inside pattern reverses.
3. For rational inequalities, include denominator zeros as forbidden boundaries.

BOTTOM LINE

Quadratic inequalities are interval questions.

Practice 02

SAME IDEA Use SIGN: Quadratic Sign Chart.

QUESTION

Solve $(x+2)(x-5) \geq 0$.

YOUR SOLUTION

03 SHADE: Graph Region

TRIGGER *region satisfies inequalities*

BECOMES Draw boundaries, then shade the wanted side.

FIRST LINE TO WRITE

$$y = mx + c$$

SIMPLEST STRATEGY

1. **S**ketch the boundary line.
2. **H**andle solid or dashed boundary.
3. **A**pply a test point.
4. **D**arken the correct side.
5. **E**xpress the region clearly.

WORKED MODEL

For $y > 2x + 1$, test $(0, 0)$: $0 > 1$ false, so shade away from the origin.

HARD VARIANTS

1. Use a dashed line for strict inequalities and a solid line for inclusive ones.
2. For several inequalities, shade one at a time and keep only the overlap.
3. If the boundary is curved, test a point after drawing the curve.

BOTTOM LINE

A test point prevents shading the wrong side.

Practice 03

SAME IDEA Use SHADE: Graph Region.

QUESTION

> Show the region $y \leq x + 3$ on a sketch.

YOUR SOLUTION

04 RANGE: Read The Range

TRIGGER *find possible values*

BECOMES Turn the expression into a boundary inequality.

FIRST LINE TO WRITE

> boundary first, then direction

SIMPLEST STRATEGY

1. **R**ewrite the condition.
2. **A**nalyse the boundary.
3. **N**ote excluded values.
4. **G**ive interval notation if needed.
5. **E**ndpoints checked.

WORKED MODEL

$$x^2 \geq 0 \Rightarrow x^2 + 3 \geq 3.$$

HARD VARIANTS

1. Complete the square or use a graph before writing the range.
2. Check whether endpoints are included from the domain and the inequality sign.
3. For transformed functions, move the range after moving the graph.

BOTTOM LINE

Always ask whether the endpoint is included.

Practice 04

SAME IDEA Use RANGE: Read The Range.

QUESTION

Find the range of $x^2 - 4$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
solve an inequality with brackets or fractions	$ax + b < c$
quadratic inequality	$ax^2 + bx + c = 0$
region satisfies inequalities	$y = mx + c$
find possible values	boundary first, then direction